

A decision under uncertainty selects an alternative A_i before a state S_j is known. Each pair (A_i, S_j) determines a payoff

$$P_{ay}(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j),$$

placed in row A_i , column S_j . The objective is to construct the complete payoff table.

Revenue and cost are

$$R = q(A_i, S_j)[r(A_i) + r(S_j)] + f_r(A_i) + f_r(S_j),$$

$$C = q(A_i, S_j)[c(A_i) + c(S_j)] + f_c(A_i) + f_c(S_j),$$

$$P_{ay}(A_i, S_j) = q(A_i, S_j)[r(A_i) + r(S_j) - c(A_i) - c(S_j)] + f_r(A_i) + f_r(S_j) - f_c(A_i) - f_c(S_j).$$

Variable components are per-unit rates multiplied by $q(A_i, S_j)$; fixed components are not. A_i , S_j , and (A_i, S_j) indicate dependence on the alternative, state, or both. Missing components are zero; directly given R or C require no decomposition.

Context Descriptions

For each problem, **(C1)** *identify the decision alternatives, the states of nature, the uncertain event, and the consequence.* Then **(C3)** *calculate revenue, cost, and profit for every alternative–state pair and construct the payoff table.*

1. **Direct case.** A small café must decide whether to launch an artisan-dessert line or keep the current menu before foot traffic is known. Traffic may be high or low. All values are in thousand USD. Launching the dessert line produces revenue and cost of 58 and 16 under high traffic, and 27 and 19 under low traffic. Keeping the current menu produces revenue and cost of 49 and 21 under high traffic, and 41 and 21 under low traffic.
2. **One-component problem.** A custom-printing cooperative must choose between producing premium notebooks and producing canvas bags before market conditions are known. Conditions may be strong demand with stable material prices or weak demand with higher material prices. The respective quantities are 6 and 4 production lots for premium notebooks, and 5 and 3 lots for canvas bags. Revenue per lot depends on the product: 12 thousand USD for premium notebooks and 15 thousand USD for canvas bags. Cost per lot depends on the state: 5 thousand USD under strong demand and stable prices, and 8 thousand USD under weak demand and higher prices.
3. **Two-component problem.** A farm cooperative must choose between a standard produce box and a premium produce box before seasonal demand is known. Demand may be strong or weak. Quantities, in hundreds of subscriptions, are 7 and 4 for the standard box and 6 and 3 for the premium box under strong and weak demand, respectively. The plan-based revenue rates are 8 and 11 thousand USD per hundred subscriptions for the standard and premium boxes. Strong demand adds a state-based revenue rate of 3 thousand USD per hundred subscriptions; weak demand adds 1. The plan-based cost rates are 4 and 6 thousand USD per hundred subscriptions, respectively. The fixed distribution costs are 9 thousand USD under strong demand and 13 thousand USD under weak demand.
4. **Three-component problem.** A regional logistics company must choose a standard or express delivery plan before demand and traffic conditions are known. Conditions may be high demand with light traffic or moderate demand with heavy traffic. Quantities, in hundreds of delivery batches, are 8 and 5 for the standard plan and 10 and 6 for the express plan under the two states, respectively. The plan-based revenue rates are 9 and 11 thousand USD per hundred batches; the state-based revenue rates are 3 and 2; and the plan-based fixed revenues are 16 and 24 thousand USD. The plan-based cost rates are 4 and 5 thousand USD per hundred batches; the state-based cost rates are 1 and 3; and the state-based fixed costs are 10 and 15 thousand USD.

1 Solutions: Revenue, Costs, Profit, and Payoff Tables

Problems:

- 1.1. Local Café Investment (C1–C3)

1.2. Custom Printing Product Line (C1–C3)

1.3. Farm Produce Subscription Plans (C1–C3)

1.4. Regional Parcel Delivery (C1–C3)
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1.1 Exercise 1: Local café investment (direct case)

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A1: Launch dessert line	A2: Keep current menu
States of nature	S1: High traffic	S2: Low traffic
Events	Actual foot traffic after the menu decision	
Consequences	Profit in thousand USD for each alternative–state pair	



C3 - Calculating profit and building the payoff table. The total revenue and total cost are supplied directly for every cell, so

$$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j).$$

Calculations (thousand USD).

Alternative and state	Revenue	Cost	Profit
Dessert line–high traffic	58	16	$58 - 16 = 42$
Dessert line–low traffic	27	19	$27 - 19 = 8$
Current menu–high traffic	49	21	$49 - 21 = 28$
Current menu–low traffic	41	21	$41 - 21 = 20$

Final payoff table (profit in thousand USD).

Alternative	High traffic	Low traffic
Launch dessert line	42	8
Keep current menu	28	20

1.2 Exercise 2: Custom printing product line (one component)

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A1: Produce premium notebooks	A2: Produce canvas bags
States of nature	S1: Strong demand and stable material prices	S2: Weak demand and higher material prices
Events	Market conditions revealed after the product-line decision	
Consequences	Profit in thousand USD for each product-line–market-state pair	



C3 - Calculating profit and building the payoff table. Revenue has one alternative-based variable component, while cost has one state-based variable component:

$$R = q(A_i, S_j)r(A_i), \quad C = q(A_i, S_j)c(S_j), \quad P = R - C.$$

Calculations (thousand USD).

Cell	Revenue	Cost	Profit
(A_1, S_1)	$6(12) = 72$	$6(5) = 30$	$72 - 30 = 42$
(A_1, S_2)	$4(12) = 48$	$4(8) = 32$	$48 - 32 = 16$
(A_2, S_1)	$5(15) = 75$	$5(5) = 25$	$75 - 25 = 50$
(A_2, S_2)	$3(15) = 45$	$3(8) = 24$	$45 - 24 = 21$

Final payoff table (profit in thousand USD).

Alternative	Strong demand, stable prices	Weak demand, higher prices
Premium notebooks	42	16
Canvas bags	50	21

1.3 Exercise 3: Farm produce subscription plans (two components)

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A1: Offer a standard produce box	A2: Offer a premium produce box
States of nature	S1: Strong seasonal demand	S2: Weak seasonal demand
Events	Seasonal market conditions revealed after the subscription-plan decision	
Consequences	Profit in thousand USD for each plan–market-state pair	



C3 - Calculating profit and building the payoff table. Revenue combines alternative-based and state-based variable components. Cost combines an alternative-based variable component and a state-based fixed component:

$$R = q(A_i, S_j)[r(A_i) + r(S_j)], \quad C = q(A_i, S_j)c(A_i) + f_c(S_j), \quad P = R - C.$$

Calculations (thousand USD).

Cell	Revenue	Cost	Profit
(A_1, S_1)	$7(8 + 3) = 77$	$7(4) + 9 = 37$	$77 - 37 = 40$
(A_1, S_2)	$4(8 + 1) = 36$	$4(4) + 13 = 29$	$36 - 29 = 7$
(A_2, S_1)	$6(11 + 3) = 84$	$6(6) + 9 = 45$	$84 - 45 = 39$
(A_2, S_2)	$3(11 + 1) = 36$	$3(6) + 13 = 31$	$36 - 31 = 5$

Final payoff table (profit in thousand USD).

Alternative	Strong seasonal demand	Weak seasonal demand
Standard produce box	40	7
Premium produce box	39	5

1.4 Exercise 4: Regional parcel delivery (three components)

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A1: Use the standard delivery plan	A2: Use the express delivery plan
States of nature	S1: High demand with light traffic	S2: Moderate demand with heavy traffic
Events	Demand and traffic conditions revealed after the delivery-plan decision	
Consequences	Profit in thousand USD for each delivery-plan–state pair	



C3 - Calculating profit and building the payoff table. Revenue has alternative- and state-based variable components and an alternative-based fixed component. Cost has alternative- and state-based variable components and a state-based fixed component:

$$\begin{aligned} R &= q(A_i, S_j)[r(A_i) + r(S_j)] + f_r(A_i), \\ C &= q(A_i, S_j)[c(A_i) + c(S_j)] + f_c(S_j), \\ P &= R - C. \end{aligned}$$

Calculations (thousand USD).

Cell	Revenue	Cost	Profit
(A_1, S_1)	$8(9 + 3) + 16 = 112$	$8(4 + 1) + 10 = 50$	$112 - 50 = 62$
(A_1, S_2)	$5(9 + 2) + 16 = 71$	$5(4 + 3) + 15 = 50$	$71 - 50 = 21$
(A_2, S_1)	$10(11 + 3) + 24 = 164$	$10(5 + 1) + 10 = 70$	$164 - 70 = 94$
(A_2, S_2)	$6(11 + 2) + 24 = 102$	$6(5 + 3) + 15 = 63$	$102 - 63 = 39$

Final payoff table (profit in thousand USD).

Alternative	High demand, light traffic	Moderate demand, heavy traffic
Standard delivery plan	62	21
Express delivery plan	94	39